FAANG –->MAANG
Why Typescript ?
TypeScript is essentially **JavaScript + Static Typing + OOP + Tooling** which makes it suitable for **large scalable enterprise applications**.

1. Why did Microsoft come up with Typescript lang ?
    a. Windows : any layman user, IT professional
    b. Office
    c. Azure

Reason	Explanation
Lack of Type Safety	JavaScript is dynamically typed. Errors occur at runtime instead of compile time. TypeScript introduces **static typing**.
Maintainability Issues	Large JavaScript projects become difficult to manage. TypeScript improves **code structure and readability**.
Better Tooling	TypeScript provides **auto-completion, navigation, refactoring, and IntelliSense** in IDEs.
Scalability	Large enterprise applications need strong typing and modular architecture.
OOP Support	TypeScript provides **classes, interfaces, generics, decorators**, making it closer to enterprise languages like C# or Java.
Compile-time Error Detection	Errors can be caught during compilation instead of runtime.

2. What are the templates to get started with Typescript in Visual Studio ?
3. What type of profile required Typescript skills ?
4. Getting started with typescript basics and advanced typescript !

Since Microsoft has C# -
C → Dennis Ritchie
C with classes
C++ -> Bjarne Stroustrup

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Objective C - I OS Mobile app dev
C# - Anders Heijlsberg _ Working in MS
Is focussed for enterprise applications like typescript

Introduction & Core TypeScript Concepts	Introduction to TypeScript and its benefits over JavaScript Setting up the TypeScript development environment(CLI/ VScode/VSstudio) Creating and managing TypeScript projects Core TypeScript concepts (types, interfaces, functions, classes, modules) Type safety and compile-time error checking
Tooling, Build & Execution	Using Visual Studio / VS Code features for TypeScript development Project configuration using tsconfig.json Building and transpiling TypeScript applications Running and debugging TypeScript applications Best practices for structuring scalable TypeScript projects

function add(a, b){
return a + b;
}
 	
add("10", 20); // unexpected result
	function add(a: number, b: number): number {
	return a + b;
	}
	
	add("10",20); // Compile-time error

Visual Studio Project Templates

Template	Purpose
ASP.NET Core Web Application + TypeScript	Backend + frontend TypeScript apps
Angular Project	Uses TypeScript by default
React + TypeScript	React application with TypeScript
Node.js TypeScript Project	Backend API development
Class Library with TypeScript	Reusable libraries
Empty Web Project + TypeScript	Custom setup

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CLI Templates (Common Modern Approach)

Framework	Command
Angular	`ng new app-name`
React	`npx create-react-app app-name --template typescript`
Vite	`npm create vite@latest app-name -- --template react-ts`
Node + TS	`npm init -y` + `tsc --init`

Job Role	Why TypeScript is Used
Frontend Developer	Frameworks like Angular and React
Full Stack Developer	Node.js backend + frontend
Angular Developer	Angular is built entirely using TypeScript
React Developer	Increasing adoption for large projects
Software Engineer	Enterprise web applications
UI Engineer	Scalable frontend architecture
Cloud Application Developer	Serverless and microservices
Technical Architect	Designing scalable frontend frameworks for enterprise App development

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```
You can learn more at https://aka.ms/tsconfig

C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI>dir
 Volume in drive C has no label.
 Volume Serial Number is CEF1-676E

 Directory of C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI

07-03-2026  11:14    <DIR>          .
07-03-2026  11:13    <DIR>          ..
07-03-2026  11:13               252 package.json
07-03-2026  11:14             1,164 tsconfig.json
               2 File(s)          1,416 bytes
               2 Dir(s)  335,684,018,176 bytes free

C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI>app.ts
'app.ts' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI>code .

C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI>dir
 Volume in drive C has no label.
 Volume Serial Number is CEF1-676E

 Directory of C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI

07-03-2026  11:15    <DIR>          .
07-03-2026  11:13    <DIR>          ..
07-03-2026  11:16               104 app.ts
07-03-2026  11:13               252 package.json
07-03-2026  11:14             1,164 tsconfig.json
               3 File(s)          1,520 bytes
               2 Dir(s)  335,546,834,944 bytes free

C:\Users\Parth Shukla\Wipro NGA MS-Dynamics_C3\TScript_demo_CLI>
```

TypeScript Basics

Concept	Description	Example
Variables	Define with types	`let age:number=25`
Data Types	number, string, boolean	`let name:string="Parth"`

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Arrays	Typed arrays	`let nums:number[]=[1,2,3]`
Functions	Type-safe functions	`function add(a:number,b:number)`
Interfaces	Define structure	`interface User{ id:number }`
Classes	OOP support	`class Student{}`
Enums	Named constants	`enum Role{Admin,User}`

Intermediate TypeScript

Feature	Purpose	Example
Union Types	Multiple types	`string
Type Aliases	Custom types	`type ID = number
Optional Properties	Optional values	`name?:string`
Function Types	Typed functions	`(a:number)=>number`
Modules	Import export	`export class User{}`

Advanced TypeScript

Feature	Use Case
Generics	Reusable components
Decorators	Angular framework
Utility Types	Partial, Pick, Omit
Type Guards	Runtime type checking
Conditional Types	Dynamic types
Mapped Types	Transform existing types

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Advanced Generics	Framework-level coding

Real World Usage - TypeScript is heavily used with:

Technology	Role
Angular	Primary language
React	Frontend development
Node.js	Backend APIs
NestJS	Enterprise Node framework
Electron	Desktop apps
Next.js	Full stack React apps

Case Study Assignment : Mini Project for github

CLI Based Student Course Enrollment System

Scenario

A training organization wants to build a small **command-line application** using TypeScript running on Node.js.

The goal is to maintain a **basic student enrollment system** where administrators can:

- Add students

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- Store student information
- Assign a role to users
- View student records

The development team will implement this system using **TypeScript features such as data types, interfaces, enums, classes, arrays, union types, optional properties, and modules**.

Learners will build this system using the **TypeScript CLI (`tsc`) compiler**.

User Story ID	User Story	TypeScript Concept
US1	As an admin, I want to store student details using strongly typed variables	Variables and Data Types
US2	As a system, I want to define a standard structure for student objects	Interfaces
US3	As a system, I want to define different user roles	Enums
US4	As an admin, I want to maintain multiple students	Arrays
US5	As a developer, I want to implement a class to represent a student	Classes

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US6	As a system, I want student IDs to accept either numbers or text	Union Types
US7	As an admin, I want email to be optional for students	Optional Properties
US8	As a developer, I want reusable custom types	Type Aliases
US9	As a developer, I want to separate logic using modules	Modules
US10	As a system, I want functions to add and retrieve students	Functions

Task 1 – Create Project	
Task 2 – Create Project Structure	
Task 3 – Define User Roles	
Task 4 – Create Student Interface	
Task 5 – Create Custom Type	
Task 6 – Implement Student Class	
Task 7 – Create Student Management Service	Use `Student[]` Typed array

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	Create methods `addStudent()` and `getStudents()`
Task 8 – Implement Application Logic	
Task 9 – Compile the Application	

Criteria	Beginner	Intermediate	Advanced
Project setup	Requires assistance	Partially independent	Fully independent
Interface design	Basic fields	Correct typing	Optimized structure
Class implementation	Minimal logic	Constructor used	Proper OOP design
Functions	Limited functionality	Correct implementation	Clean modular functions
TypeScript usage	Few concepts	Multiple concepts	Most concepts applied

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